

Electronics

3 BYJ stepper motors

<https://robocraze.com/products/5v-stepper-motor?variant=40193680507033&country=IN¤cy=INR>

Convert the motors from Uni Polar to Bi polar by following this tutorial with this the motors become more precise

<https://www.youtube.com/watch?v=pOMRzBdshEY&t=94s>

3 TMC 2208 stepper drivers <https://robu.in/product/tmc2208-stepper-motor-driver-module/>

3 Stepper driver expansion boards

[obu.in/product/red-a4988-stepper-motor-driver-controller-board/?gad_source=1&gad_campaignid=20381096599&gbraid=0AAAAADvLFWeWZweoldbgVLavXCJuIP1wi](https://robu.in/product/red-a4988-stepper-motor-driver-controller-board/?gad_source=1&gad_campaignid=20381096599&gbraid=0AAAAADvLFWeWZweoldbgVLavXCJuIP1wi)

12V 2A DC power supply

https://www.amazon.in/Electronic-Spices-Supply-Adapter-Connector/dp/B0CWTQWN3F?source=ps-sl-shoppingads-lpcontext&ref_=fplfs&psc=1&smid=A1UUD7CBBVVXIG

Arduino Nano

https://robocraze.com/products/nano-development-board-compatible-with-arduino?_pos=1&_sid=7fe53113e&_ss=r

WS28182 Ring Light :

<https://www.flyrobo.in/7-bit-ws2812-5050-rgb-led-built-full-color-driver-colored-light-round?tracking=ads> (the one I used is no longer available , you can use any light source btw)

Jumper wires and a barrel jack connector for power

3D Printed Parts and screws

(Print with PETG only)

Main Body:  ThreadScope 3D parts

Motor couplers x3:  ThreadScope 3D parts

3 M3 hex screws and nuts ([OnlyScrews.com](https://www.OnlyScrews.com) got good sets)

